ACCEPTANCE SAMPLING

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Acceptance Sampling

- Acceptance sampling involves evaluating a portion of the product in a lot for the purpose of accepting or rejecting the entire lot as either confirming or not confirming to quality specifications. For example, while purchasing food grains, the usual procedure is to take a handful of the grains in our hands and inspecting its quality on the basis of size, color, odor and texture.
- If that handful of grains is found good we decide to purchase the grains. However, if we are not satisfied, we inspect another handful of a different grain and repeat the procedure.
- The "handful of grains" represents the *sample* on which inspection is carried to determine its acceptability. The different types of grains represent the *lot*.

Acceptance Sampling by Attributes

- Here, "attributes" refer to characteristics that are either present or absent, or that can be classified as conforming or non-conforming, rather than being measured on a continuous scale.
- The statistical method used to decide the acceptance or rejection of lots is provided by a curve. This curve called operating characteristic (O.C.) curve is a graph of fraction defective in a lot against the probability of acceptance.
- N = lot size
- n = sample size
- c = number of defective items in the sample permitted
- P_A = The probability of acceptance
- p = The fraction or percent defective
- The variable c is called the *acceptance number* of defectives. Thus the value of c = 2 means that if number of defectives are 0 (no defective), 1 or 2, the lot would be considered acceptable. If more than two defectives are found the lot would be rejected.

Construction of OC curve

Let N = 1000, n = 60, c = 3 and p = 0.01
The above sampling plan means that one should take a random sample of 60 from a lot of 1000 items.
If the sample contains more than 3 defectives, reject the lot otherwise accept it.
The value of probability of acceptance (P_A) is calculated using the Poisson's formula:

$$P_A(c) = \frac{(n \times p)^c \, e^{-(n \times p)}}{c!}$$

Percent defective in a sample of size 60 is 0.01, and the value *c* is 3, the probability of its acceptance:

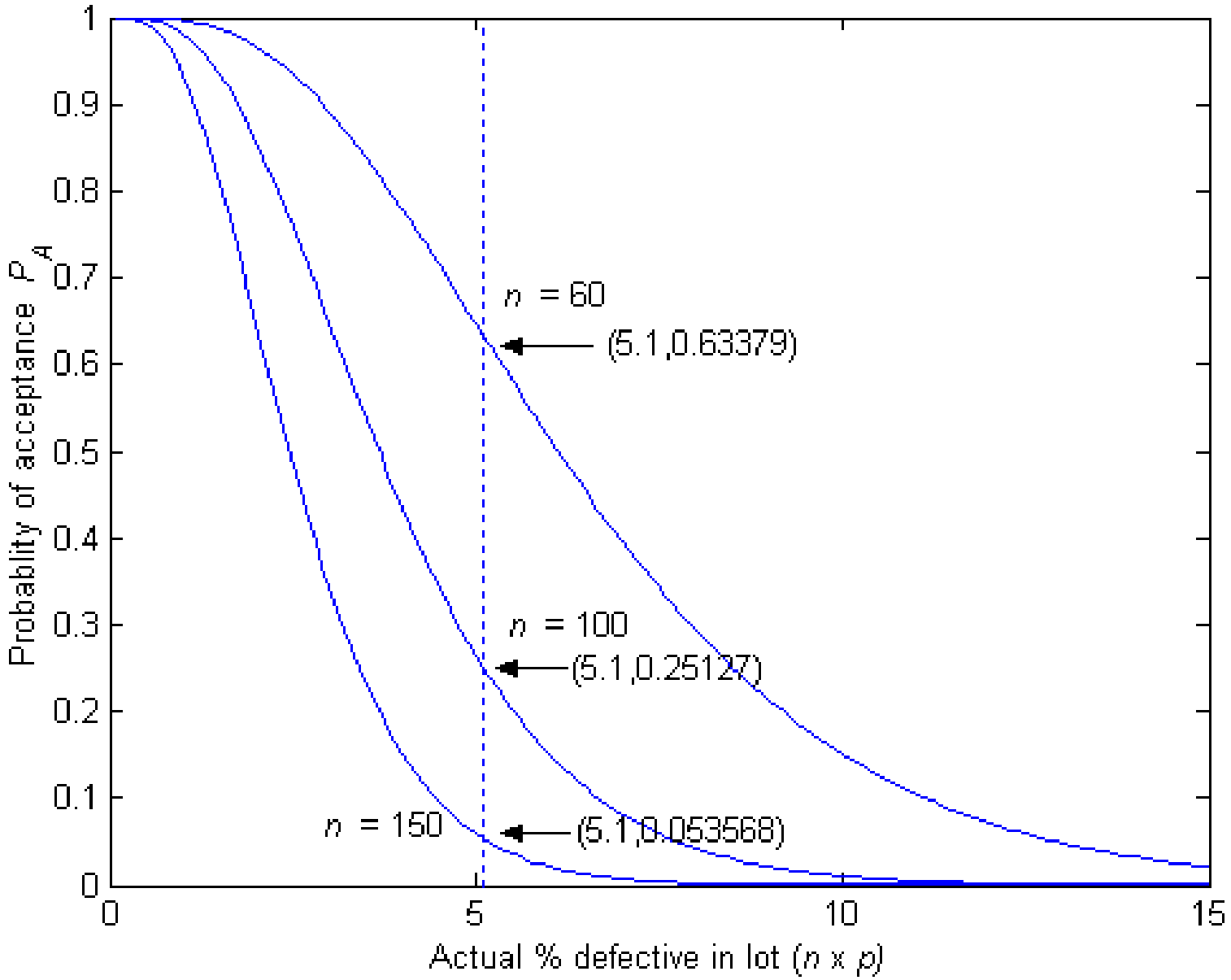
$$P_A(3) = \frac{(60 \times 0.01)^3 e^{-(60 \times 0.01)}}{3!} = 0.0198$$

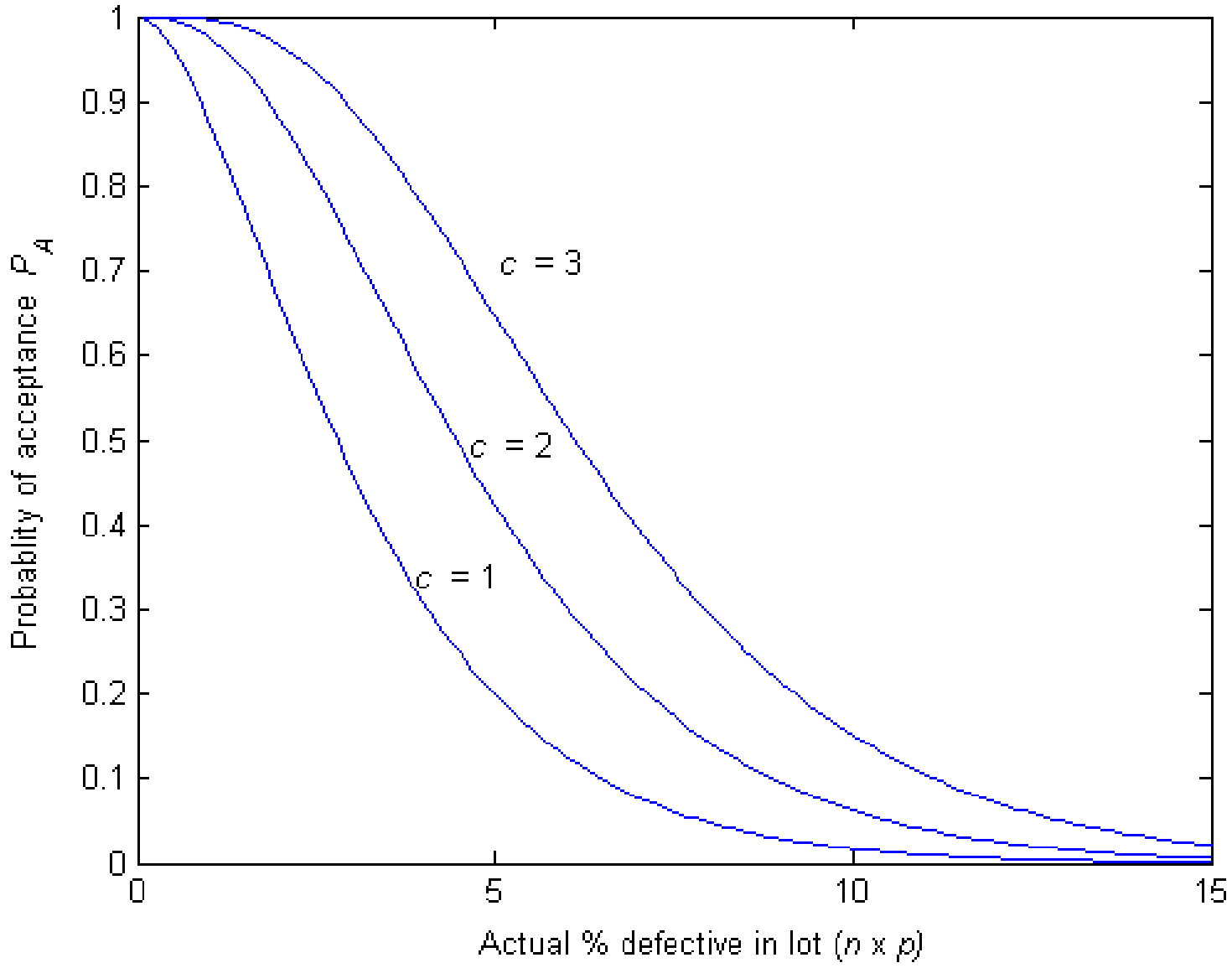
$$P_A(2) = 0.0988, P_A(1) = 0.3293 \text{ and } P_A(0) = 0.5488$$

$$P_A(3) + P_A(2) + P_A(1) + P_A(0)$$

$$= 0.0198 + 0.0988 + 0.3293 + 0.5488 = 0.9967$$

% defective (p)	n×p	P_A
1	60×0.01 = 0.6	0.9967
2	60×0.02 = 1.2	0.9662
4	60×0.04 = 2.4	0.7784
5	60×0.05 = 3	0.6472
7	60×0.07 = 4.2	0.3955
8	60×0.08 = 4.8	0.2942
10	60×0.1 = 6	0.1512
12	60×0.12 = 7.2	0.0719
15	60×0.15 = 9	0.0212


1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
Probablity of acceptance P_A
0
5
10
15
Actual % defective in lot (n x p)
n = 60
(5.1,0.63379)
n = 100
(5.1,0.25127)
n = 150
(5.1,0.053568)


Probablity of acceptance P_A
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
5
10
15
Actual % defective in lot (n x p)
c = 3
c = 2
c = 1

Producer's and Consumer's Risk

- A producer always tries to produce the products of high quality in the hope that such a lot is accepted by the customer.
- The acceptable quality level (AQL) is regarded as a good level of quality.
- AQL represents the maximum proportion of defectives which the consumer finds definitely acceptable.
- However, an ideal sampling plan that discriminates the good lots perfectly always from bad lots is rare.
- Hence, there always remains a risk that a good lot will not be accepted.
- The *producer's risk* (denoted by α) is defined as the probability that lots of *good* quality level AQL will *not* be accepted.
- For example, α = 0.05 means that in the long run about 1 in 20 lots will be rejected even though the lots are coming from a process controlled at AQL level.
- Consumer risk, also known as type II error or beta risk, is the probability of a bad product being mistakenly accepted and reaching the consumer, potentially leading to financial, reputational, or even safety issues

Consumer's risk

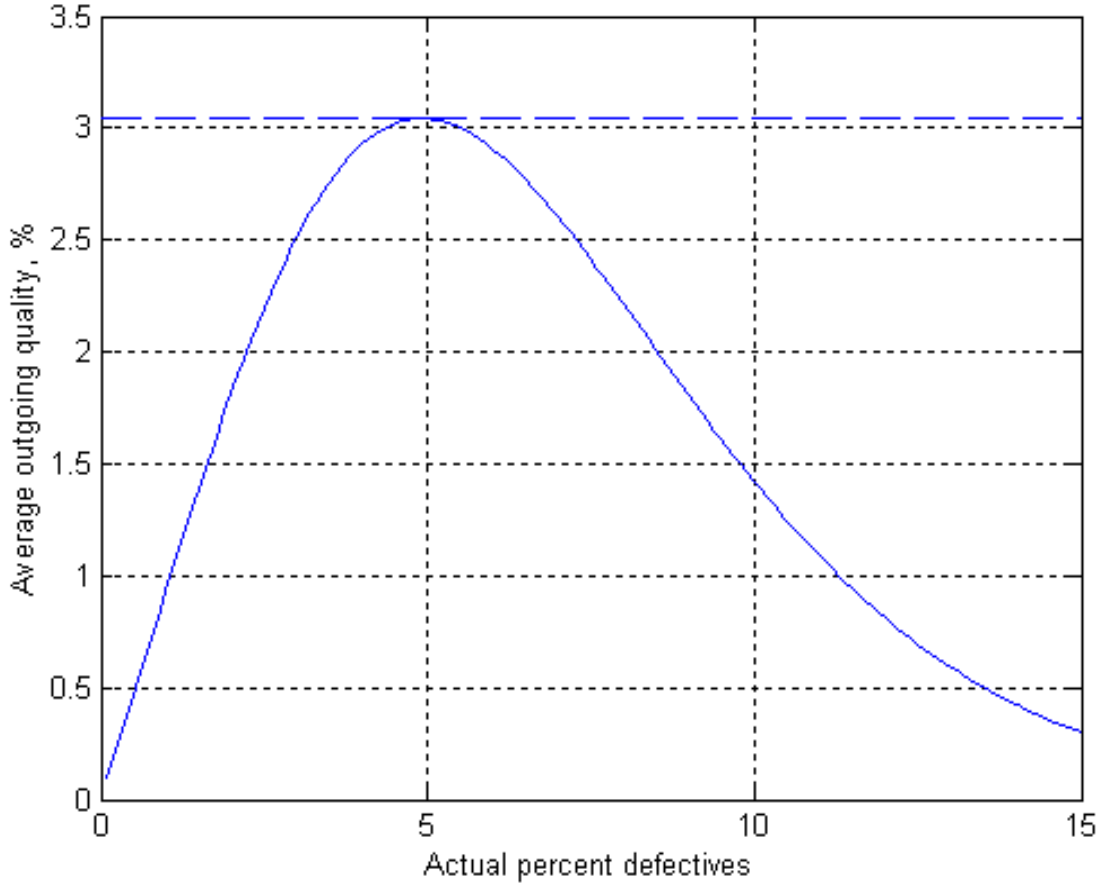
- Lot tolerance percent defective (LTPD) is the dividing between good and bad lots.
- Since lots of this quality level is regarded as poor, it is expected that the probability of *accepting* such a lot is *low*.
- However, if the consumer accepts such a lot entire loss has to be borne by him.
- The *consumer's risk* (denoted by β) is the probability that lots of quality level LTPD will be accepted which otherwise should be rejected.
- If $\beta = 0.1$ and LTPD =2.5%, it means that the consumer will most likely accept 10% of the lots containing 2.5% defectives.

Average Outgoing Quality (AOQ)

A rejected lot is subjected to 100% inspection and all defectives found are replaced by good parts. This results in a lot free of any defectives. If a lot is accepted by sample then there is a risk that some defectives have passed. The AOQ represents the average percent defective in the outgoing products after inspection

- The number of defectives in the lot after n items in a sample has been passed is $p(N\text{-}n)$.
- This multiplied by the probability of acceptance provides the outgoing quality of lot *i.e.* $P_A\,(N\text{-}n)p$.
- Hence, the AOQ. is this number divided by the number of items in the lot, N.

$$\therefore AOQ = P_A p\left(\frac{N-n}{N}\right)\times 100$$



This peak in the curve is called AOQL

AOQ curve for N =1000, n = 60, c = 3

$$\therefore AOQ = P_A p\left(\frac{N-n}{N}\right)\times 100$$

Single Sampling Plan

- In such a plan only three numbers are specified, *i.e.* lot size (*N*), sample size (*n*) and acceptance number (*c*).
- For example *N* = 1000, *n* = 60 and *c* = 3 means that the sampling plan consists of 60 items from a lot of 1000; the lot is accepted if the sample contains more than 3 defectives otherwise accepted.

Double Sampling Plan

- In a double sampling plan the decision on acceptance or rejection of the lot is based on two samples. This type of decision becomes important when the sample taken for inspection is neither good nor bad enough. The procedure of a double sampling plan is illustrated through the following example.
- Let $N = 600$
- $n_1 = 30$, number of pieces in the first sample
- $c_1 = 1$, acceptance number for first sample
- $n_2 = 45$, number of pieces in the second sample
- $c_2 = 3$, acceptance number for the two samples combined together.

i) Take a first sample of 30 items from a lot of 600 and inspect.

ii) Accept the lot on the basis of first sample if it contains 0 or 1 defective.

iii)Reject the sample if it contains more than 3 defectives.

iv)If the first sample contains 2 or 3 defective take a second sample of 45 items.

v) Accept the lot on the basis of first and second sample combined if the combined sample of 75 items contains 3 or less defectives.

vi)Reject the lot on the basis of combined sample if the combined sample of 75 items contains more than 3 defectives.

a) 0 or 1 defective in the first sample without taking second sample

In order to calculate the probability of this lot assume that population is 5% defective.

Therefore, the number of defective items = 600×0.05 =30.

The probability for zero defective in the first sample is

$$P_A(0) = \frac{(30\times 0.05)^0 e^{-(30\times 0.05)}}{0!} = 0.223$$

The probability for one defective in the first sample is

$$P_A(0) = \frac{(30\times 0.05)^1 e^{-(30\times 0.05)}}{1!} = 0.335$$

Hence, the total probability of acceptance of 0 or 1 defective in the first sample without taking second sample is
0.223 + 0.335 = 0.558.

b) 2 defectives in the first sample followed by 0 or 1 defective in the second sample.

Now for second sampling plan, N = 600-30 =570. Since 2 defectives are already in the first sample, hence the number of defectives left = 30–2 = 28. Hence percent defectives = (28/570) ×100 = 4.91%. Hence, the probability

$$= \frac{(30\times 0.05)^2 e^{-(30\times 0.05)}}{2!} \times \left(\frac{(45\times 0.0491)^0 e^{-(45\times 0.0491)}}{0!} + \frac{(45\times 0.0491)^1 e^{-(45\times 0.0491)}}{1!} \right)$$

$$= 0.25102 \times (0.1097+0.2425) = \underline{0.0884}$$

3 defectives in the first sample followed by 0 defective in the second sample.

Since 3 defectives are already in the first sample, hence the number of defectives left = 30–3 = 27. Hence percent defectives = (27/570) ×100 = 4.74%. Hence, the probability is

$$= \frac{(30\times 0.05)^3 e^{-(30\times 0.05)}}{3!} \times \left(\frac{(45\times 0.0474)^0 e^{-(45\times 0.0474)}}{0!} \right)$$

= 0.1225 × 0.1184 = <u>0.0145</u>

Thus, the total probability of acceptance of this lot = 0.558 + 0.0884 + 0.0145 = 0.661

Acceptance Sampling by variables

A quality control method where **measured values** (like length, weight, or strength) of a sample are used to determine acceptance or rejection of a lot.

1. **Known σ (Standard Deviation):**

$$\bar{X} \pm k \cdot \sigma$$

Lot accepted if sample mean $\bar{X}$ is within limits.

2. **Unknown σ (Use s):**

$$\bar{X} \pm k \cdot s$$

Use sample standard deviation **s**.

Acceptance Criteria:

- Specification limits: and
- Sample data used to estimate whether the **entire lot meets the specs**.